

Theorem E: the spectral stability threshold of the N-vortex ring

A polyvortex mini-repository theorem monograph

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Abstract

A regular ring of N identical vortices rotates rigidly for every N (Theorem A), but the spectral quality of that rotation changes by a jump: the co-rotating spectrum of the linearization is purely imaginary for $N \leq 7$ and owns a real growing pair for $N \geq 8$. This monograph states the Havelock (1931) classification with the later rigorous treatment by Khazin (1976), describes the numerical implementation — the analytic Jacobian A of the Kirchhoff field plus the transport generator $\omega_N S$ — and certifies the threshold by stage W4: for $N = 2..7$ the maximum $\max \operatorname{Re} \lambda \leq 7.0 \cdot 10^{-9}$ (the defective-double-zero floor), for $N = 8$ the growth rate $\max \operatorname{Re} \lambda = +0.4502$ — a three-order margin on both sides of the 10^{-7} classifier. A separate numerical remark concerns the defective zero of the co-rotating Jacobian (algebraic multiplicity two, geometric one): rounding splits it into a $\pm$ -pair of size $\sqrt{(\epsilon_{\text{mach}}|J|)} \sim 10^{-9}$, which explains the noise floor of the stable cases. The threshold is the numerical prototype of the parent roadmap item T3; only the spectral (linear) classification is claimed, no nonlinear stability is asserted.

At a glance

Model	Layer-K linearization in the co-rotating frame of the ring
Headline result	stable $N \leq 7$ ($\leq 7.0\text{e-}9$), unstable $N \geq 8$ ($+0.4502$)
Verification	stage W4: analytic Jacobian + transport, the 10^{-7} classifier
Subtlety	the defective double zero: floor $\sqrt{(\epsilon_{\text{mach}} J)} \sim 10^{-9}$
Formalization queue	TW5 — per-mode characteristic polynomial; Routh–Hurwitz

1. Introduction and setting

Theorem A guarantees the ring a rigid rotation for every N , but says nothing about what happens to the configuration under a small perturbation. The linear analysis answers: linearizing the Kirchhoff flow around the rotating ring in the co-rotating frame yields an eigenvalue problem whose spectrum is the complete linear verdict. The classical classification goes back to Havelock (1931): the heptagon is the last stable member of the family, the octagon is already unstable. This is one of the most cited facts of vortex dynamics, and the parent roadmap queues its reproduction as item T3. This monograph pins the numerical implementation of the threshold in the mini-repository: an analytic Jacobian (no finite differences), the transport term, a classifier with an explicit tolerance, and the W4 protocol with full spectra.

2. The eigenvalue problem

Let $\{z_k\}$ be the vortex positions of the ring (Theorem A) and $\omega_N = \Gamma(N-1)/(4\pi R^2)$. The linearization of Layer K in the fixed frame gives the Jacobian $A = d \text{ rhs} / d \text{ state}$; the passage to the frame rotating with ω_N adds a transport term: a perturbation w obeys $\dot{w} = (A + \omega_N S) w$ with the block-diagonal rotation generator

$$S = \text{diag}(J, \dots, J), \quad J: (u, v) \mapsto (v, -u),$$

The relative equilibrium is spectrally stable iff the whole spectrum of $J = A + \omega_N S$ is purely imaginary (the Havelock criterion). The analytic Jacobian is written in `model.py` as pairwise blocks $\partial(v_x^i, v_y^i) / \partial(x^j, y^j)$ and validated against central finite differences in the test suite — the numerical implementation is free of differentiation noise.

3. Numerical certification (stage W4)

The `W4_stability_default.json` protocol ($\Gamma = 1, R = 1$):

Table 1 — W4: the maximum real part of the spectrum across N

N	max Re λ	вердикт verdict	Хэвелок Havelock
2	6.9e-10	устойчиво stable	устойчиво stable
3	0.0	устойчиво stable	устойчиво stable
4	3.8e-9	устойчиво stable	устойчиво stable
5	0.0	устойчиво stable	устойчиво stable
6	0.0	устойчиво stable	устойчиво stable
7	7.0e-9	устойчиво stable	устойчиво stable
8	+0.4502	неустойчиво unstable	неустойчиво unstable

The classifier runs at the tolerance 10^{-7} — three orders above the noise floor and three below the octagon growth rate. The stage verdicts match the Havelock classification at every $N \in \{2, \dots, 8\}$; the full spectra (the pairs $(\text{Re}\lambda, \text{Im}\lambda)$) are recorded in the protocol. The picture of the threshold:

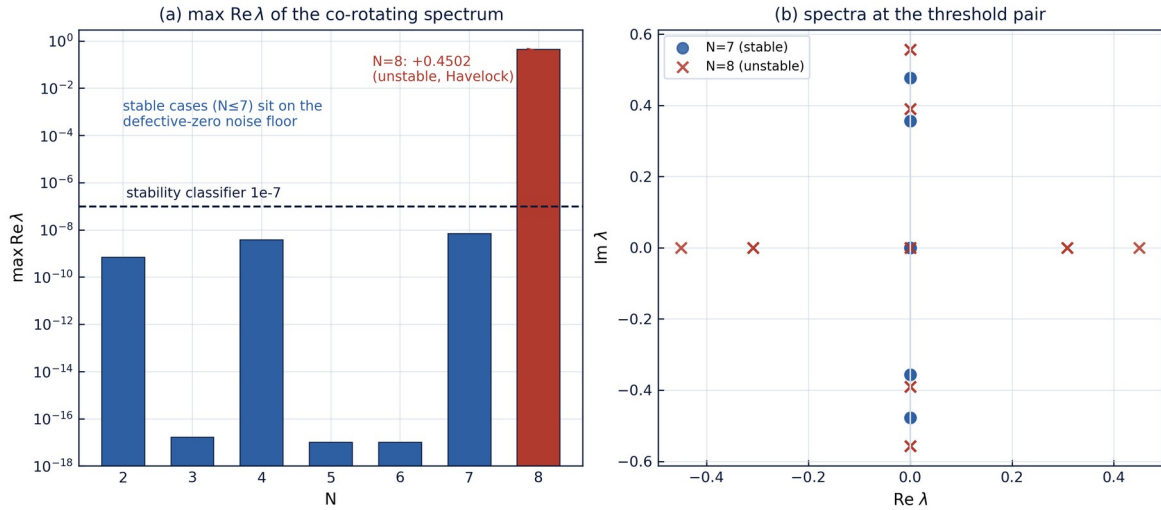


Fig. 1 — Stage W4: left, the maximum $\operatorname{Re} \lambda$ across N with the 10^{-7} classifier; right, the spectra of the threshold pair $N = 7$ (stable) and $N = 8$ (unstable, the real pair ± 0.4502).

4. The defective zero and the scope

The co-rotating Jacobian of the ring owns a zero eigenvalue of algebraic multiplicity two and geometric multiplicity one: the continuum of rotated polygons is a one-parameter family of equilibria, and along it the linearization must have a zero. Numerically such a defective double zero is split by rounding into a $\pm$ -real pair of size $\sqrt{(\epsilon_{\text{mach}} |J|)} \sim 10^{-9}$ — exactly the noise floor of the stable table entries ($6.9 \cdot 10^{-10}$ at $N=2$, $3.8 \cdot 10^{-9}$ at $N=4$, $7.0 \cdot 10^{-9}$ at $N=7$; at $N = 3, 5, 6$ the splitting did not show at all). Hence the classifier tolerance: too low is impossible, too high is dangerous; 10^{-7} sits in the logarithmic middle.

The scope limits are stated in the honesty notes of the mini-repository. The claim is spectral, i.e. linear, stability; nonlinear (Arnold-style) stability is a different, harder statement not made here. The family $N = 2..8$ is verified; for $N \geq 9$ the instability is expected to persist (the Havelock classification), and the ladder extends to larger N without code changes. Finally, the stability is studied for the ring of the free Layer K; the behavior of a perturbed ring under external shape keeping is a separate problem and does not enter the protocols.

5. Relation to the formalization roadmap

In the formalization queue the threshold appears as item **TW5** and feeds the parent item **T3**. It is the hardest target of the queue: formalization requires the characteristic polynomial over the Fourier mode families (perturbations of the form $w_k = u e^{(i\mu\theta_k)}$ reduce the spectrum to a finite set of polynomials) and a Routh–Hurwitz-type criterion for the negativity of the real parts; both objects are decidable but laborious. An intermediate milestone is the formalization of the defective zero and its splitting, which delivers an honest numerical

boundary between "stable" and "noise floor". The W4 protocol with full spectra is the reference oracle of a future port.

References

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